

OBJECTIVES OF CURRENT PAPER

BT302 - IMMUNOLOGY

Final-Term File 2024 By

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- ❖ Human can make 10^7 - 10^8 different Ab molecules
- ❖ second most common antibody: **IgA**
- ❖ How contagious is different from non contagious: A) **Contagious** diseases are spread through direct or indirect contact with an infected person, while **non-contagious** diseases cannot be spread from person to person.
- ❖ pro- inflammatory cytokines involved in up regulations of **inflammatory process**
- ❖ cell used in MHC class 2: T cells
- ❖ Role of IgE in allergic reactions: Involved in allergic reactions (Anaphylaxis)
- ❖ Primary function of coagulation: Antimicrobial activity
- ❖ How many kinds of polyadenylation: two
- ❖ Example of live attenuated vaccine: Bacille Calmette-Guerin vaccine (BCG)
- ❖ Time span of acute rejection: 1-3 weeks
- ❖ ADCC is another form of: Antibody dependent Cellular Cytotoxicity
- ❖ MASP stands for: Mannose-Associated Serine Proteases
- ❖ How many types of adjuvant ? two Type
- ❖ Second most common serum antibody is. IGA
- ❖ Humans can make different Ab molecules.: 10^7 - 10^8
- ❖ MHC 11 molecules contain highly conserved region for binding.: $\alpha 2$ & $\beta 2$
- ❖ Exudation of plasma proteins and fluids.: Edema
- ❖ MASP stands for.: Mannose-Associated Serine Proteases
- ❖ First developed antibody ---- IgM
- ❖ Cytotoxic cells produce in -----: bone marrow
- ❖ Pro- inflammatory cytokines are involved in ----- of inflammatory process __ : Up regulation
- ❖ Least common serum antibody is ____ IgE
- ❖ ICAM is stand for : Intracellular Adhesion Molecule
- ❖ Wiskott - Aldrich syndrome is a : Disorders of T-Cells(normal number of T-cells with reduced functions)
- ❖ During hyper acute rejection.....: lead graft destruction?
- ❖ There are Kind of polyadenylation? two
- ❖ In the snake venom the convert of CD3 into CD8 is Pathways.: Saponin-mediated pathways
- ❖ all of the following example of live attenuated vaccine except: Hepatitis A
- ❖ which immunological technique involves the separate addition of both antigen and antibodies into medium during experimentation:C) double immunodiffusion
- ❖ what is another name for secondary lymphoid organ of immune system: Peripheral Lymphoid Organs
- ❖ which of the following is not type of complement pathway; C) adaptive pathway
- ❖ if a complement protein deficiency results in the inability to form complexes with microorganism what might be the consequence for host defense: decreased phagocytosis
- ❖ what term is used to describe the ability of T- independent antigens to activate various clones of B-cells simultaneously:polyclonal activation
- ❖ in MHC1 molecules region is highly conserved in long Alpha chain: Alpha 3

- ❖ cytokineinhibits the cytokines production from microphage; Cytokine IL-10
- ❖ primary immunodeficiency is..... Inherited defects of the immune system
- ❖ How does the lag phase in a secondary antibody response compare to in primary response: shorter in than the primary antibody
- ❖ what is importance of adjuvant in immunological experiments based on the effects hey exert; Enhance immune response
- ❖ immunology is study of.....? immune system & immune response
- ❖ hypervariable reason and framework region are sub region of.....? variable regions.
- ❖ Ag & Ab antigen and antibody reaction which can be detected on basis of substrate utilization by enzyme is called.....? ELISA
- ❖ The process by which B- cell change the class of Ab but not specificity is name as: Class Switching
- ❖ soluble immune complex is form in: Hypersensitivity type 3
- ❖ Delayed type of hypersensitivity is.....? Type IV Hypersensitivity
- ❖)These all are true for immune evasion EXCEPT -----maximize the ability of organisms to develop infection
- ❖ Immuno- evasion occurs: either by weak immune response • Or by strategies devised by pathogens
- ❖ Cell mediated response is primary defense against..... Pathogens: interacellular
- ❖ constant occurrence of infection is? endemic
- ❖ Which of the following are poisoning: IgE
- ❖ Tolerance to is longer as compare to B- cells . (t cell)
- ❖ isograft between individual of same genetic makeup: identical twin
- ❖ Pro- inflammatory cytokines are involved in ----- of inflammatory process: Up regulation
- ❖ Third most common antigen in serum: IgA
- ❖ The third most common antibody in serum is:C) **IgM**
- ❖ Immunity Is the study of state of protection from infectious disease.
- ❖ Soluble antigens are **less immunogenic**
- ❖ Opsonin are complement proteins, igM, IgG ...All
- ❖ Poly clonal antibodies belong to....IgG
- ❖ First line of immunodeficiency..... Innate
- ❖ Resistance of body against infectious disease: Immunity
- ❖ BCR recognize THOSE AG TAHT ARE IN NATURE: soluble or free floating.
- ❖ MASP STANDS: Mannose-Associated Serine Proteases
- ❖ SOLUBEL INMUNE COMPLEX ARE : Type III Hypersensitivity
- ❖ DELAYED TYPE oF HYPERSENSITIVITY: type iv
- ❖ INFECTIONS OF BACTERIA OR SECRETORY MOLECULES OF BACTERIA LIKE TOXINS CALLED : Neutralization
- ❖ TH1 CELLS RELEASE: cytokine like IFN γ upon activation & activate macrophages
- ❖ NH OR NK CELSS ARE CALLED: Large Granular Lymphocytes (LGL)
- ❖ OPSONIS ARE: complement proteins, igM, IgG ...All
- ❖ AG WHICH INDUCE TOLERANCE: Tolerogen
- ❖ Immunity MECHANISM BIASED OF GRAFT REJECTION: 1) Time of rejection 2) Nature of allo-antigens of graft 3) Immune status of host. all
- ❖ SECONDARY IMMUNE DECFIGEFICNCY : AIDS
- ❖ AUTOANTIBODIES CAN BE AGAINST DADH KINDS OF ANTIGENS: t two kinds of antigens 1) Self cell or cell associated antigens 2) Soluble antigens
- ❖ Chimeras: animals with hybrid nature of immune cells with host's own cells & donor's cells
- ❖ PRRS ARE : Pattern recognition Receptors
- ❖ AG AND AB REACTION IS all(Non-Covalent bonding • Hydrogen bonds • Vander Wall forces • Electrostatic forces • Reversibility)
- ❖ SOMATIC HYPER MATURATION INCREASE: binding affinity of BCR for its cognate ligand
- ❖ ALPHA AND BETA CHAINS ARE CONNECTED: to form class II MHC
- ❖ COMPLEMENT PROTEINS ARE: good opsonins
- ❖ IL4 INDUCED

- ❖ Class switching of immunoglobulin occurs from IgM to ----- IgG
- ❖ ACTIVATION OF ADH ENZYMES
- ❖ IMMUNOGLOBULIN CLEAVE THE PRO INFLAMMATION OF : Up regulation
- ❖ EXAMPLE OF LIVE ATTENUATED VACCINE: polio
- ❖ ALFA FETO PRO LEVEL INCREASES during hepatocellular carcinoma
- ❖ WHAT REACT ACTIVATION COMPLEMENT DURING INCOMPLETE BLOOD TRANSFUSION : acute hemolytic
- ❖ PRIMARY FUNCTION OF THYMUS SUPPOSE 10,000 LYMPHOCYTES RECIRCULATING AND 100 ARE ENCOUNTER GIVE PERCENTAGE DEFINE: 1%
- ❖ DIRECTLY STIMULATE B CELLS : T-Independent Antigens:
- ❖ WHAT IS NOT TYPE OF A COMPLEX PATHWAY
- ❖ Soluble Antigens Less immunogenic.
- ❖ Both antibodies and T-cells are involved in autoimmunity.
- ❖ The cytokines produced from T-helper 1 (Th1) cells especially IL-2 & INF- γ .
- ❖ Least common serum antibody is IgE.
- ❖ How many kinds of polyadenylation two.
- ❖ Pro- inflammatory cytokines are involved in Up regulation of inflammatory process.
- ❖ Global occurrence infection pandemic.
- ❖ ELISA Enzyme linked immunosorbent assay.
- ❖ Infectivity of bacteria or secreted molecules of bacteria like toxins. Neutralization.
- ❖ Acute rejection time span: 1-3 weeks.
- ❖ Secondary Immunodeficiency Associated with infections e.g AIDS
- ❖ Associated with aging: hypo-cellularity
- ❖ Associated with malignancies e.g in case of Leukemia, Myeloma. All of these.
- ❖ Wiskott-Aldrich syndrome: normal number of T-cells with reduced functions.
- ❖ Resistance of body against infection: immunity
- ❖ MHC II present: Ag
- ❖ Types of Adjuvant two
- ❖ Least common serum antibody : IgE
- ❖ IgM exist in form: pentamer generally as secretory form
- ❖ Newly synthesized antibodies combine with Ag to form Ag/Ab (Immune) complexes
- ❖ Humans make 10^7 - 10^8 Ab molecules
- ❖ Cytotoxic T cells are associated with: T-Lymphocytes or Lymphoid Lineage
- ❖ The class II MHC is encoded by genes for Human MHC which are located at chromosome 6
- ❖ In T cell activation process lack of co stimulus lead to: anergy
- ❖ CD 2 Leucocyte Function of antigen : Antigen 3 (LFA-3)
- ❖ Complement proteins are produced by: liver
- ❖ Enhancement of process of phagocytosis by phagocytes
- ❖ Host vs graft rejection is: Reaction of host against allo-antigens of graft
- ❖ Commonest form of immunoglobulin deficiency: IgA deficiency
- ❖ T cell maturation in: Thymus
- ❖ cytidine deaminase present in: DNA deamination
- ❖ Sugar- antigens have the ability to activate ----- of T –cells polyclonally 25%
- ❖ Scid stand for: Severe combined immunodeficiency
- ❖ The gene for chain is located on chromosomes 14: Heavy chain
- ❖ Particulate nature of antigen causes reaction when react with corresponding antibody : Agglutination
- ❖ RSS are sequence which is flanked b/w v d and j
- ❖ Chronic rejection is characterized by with loss of graft function/ Loss of immunological tolerance by grafted tissue
- ❖ During myeloperoxidase depends intracellular killing of microbes
- ❖ Vascular pathogens are presented in combination with class II mhc
- ❖ Global occurrence infection ...pandemic
- ❖ C4a ,C3a,C5a are : Potent anaphylatoxin
- ❖ ADCC another form... Type II hypersensitivity
- ❖ immune response independent of antibody called.... Cell mediated immunity
- ❖ Toxin production is neutralized by antibodies name: (antitoxins)

- ❖ induction of active humoral immunity.. Passive Immunization
- ❖ Adjuvant have Type : 2
- ❖ components form immunoglobulin deficiency is.... IgA deficiency
- ❖ IL-12 is cytokine of.... Inflammation
- ❖ pneumococcus polysaccharide serve ... Antigen : T independent Antigen
- ❖ Alpha -Feto protein level increasing during: hepatocellular carcinoma
- ❖ Molecule mass ranges that tends to make less substances poor immunogens.
- ❖ Primary role of IgE allergic reaction... Involved in allergic reactions
- ❖ classic pathways for complement activation innitate . antibody dependent complement activation.
- ❖ MHC I present... Ag
- ❖ MHC II present...Ag
- ❖ Types of Adjuvant...two
- ❖ IgM exist in..... form: pentamer generally as secretory form
- ❖ Humans can make 10^7 - 10^8 different Ab molecules
- ❖ Monoclonal Ab used for... tumor antigen
- ❖ CD2 Leucocyte Function of which Antigen.... : Antigen 3 (LFA-3)
- ❖ types of adjuvant..... 2
- ❖ heavy chain of antibody... 14 chromosomes
- ❖ class switching into.... Ig E
- ❖ IgM and Ig G are components of antibodies
- ❖ isograft between the individual of cell: Twins
- ❖ Antigen binding groove of Class 1 MHC can accommodate the peptides consisting of ----
--(• Composed of $\alpha 1$ & $\alpha 2$ domain • Antigenic peptide reside within the Ag binding groove • Within groove peptide make contact with residue peptide (highly polymorphic) • 8-10 amino acids can accommodate)
- ❖ hypotensive type I is also called__ Immediate or Anaphylactic reactions
- ❖ IgM or cold agglutins- active at ____ 4°C
- ❖ AIDS caused by(HIV)
- ❖ acute rejection are....(1st set of rejection)
- ❖ isograft between individual of same genetic makeup ..(identical twin)
- ❖ tolerance to ..(t cell).... is longer as compared to b cell
- ❖ This state of immunodeficiency is associated with infections for example
- ❖ AIDS. There is also association of aging factor like hypo-cellularity in bone marrow.
- ❖ Malignancies for example Leukemia and Myeloma are also linked with secondary immunodeficiency.
- ❖ Animals with hybrid nature of Immune cells with host and donor cells are__Chimeras✓
- ❖ Patient suffering AIDS are susceptible to__Kaposi Sarcoma✓
- ❖ B-cells are activated directly without the help of T-cells for antibodies production in__T-independent antigens✓
- ❖ Infectious diseases that spread from person to person are said to be __Contagious✓
- ❖ T- Dependent antigens contain ----- type of epitopes with few copies__ Same✓
- ❖ Autoimmune disorders__abnormal response against self-tissues✓
- ❖ Type 1 hypersensitivity is mediated by__IgE✓
- ❖ Pro- inflammatory cytokines are involved in ----- of inflammatory process. Up regulation✓
- ❖ These all are true for immune evasion EXCEPT -----maximize the ability of organisms to develop infection✓
- ❖ These all are true for inflammation EXCEPT ---- Expressed in myeloid cells
- ❖ Adjuvant are of __ types 2
- ❖ **Immunology** is a study of **the immune system, its functions, and disorders.**
- ❖ Global occurrence infection ... **Pandemic**
- ❖ C4a ,C3a,C5a are **Potent anaphylatoxin**
- ❖ ADCC another form... **Type II hypersensitivity**
- ❖ Immune response independent of antibody called.... **Cell mediated immunity**
- ❖ Toxin production is neutralized by antibodies name (**antitoxins**)

- ❖ Induction of active humoral immunity___ **Passive Immunization**
- ❖ Adjuvant have **two Types**
- ❖ components form immunoglobulin deficiency is.... **IgA deficiency**
- ❖ IL-12 is cytokine of.... **Inflammation**
- ❖ pneumococcus polysaccharide serve ... **T independent Antigen**
- ❖ Alpha -Feto protein level increasing during ___ **hepatocellular carcinoma**
- ❖ Molecule mass ranges that tends to make less substances poor immunogens.
- ❖ Primary role of IgE___ **Involved in allergic reactions**
- ❖ classic pathways for complement activation innitate also called ___ **antibody dependent complement activation.**
- ❖ AIDS caused by(**HIV**)
- ❖ acute rejection are....(**1st set of rejection**)
- ❖ isograft between individual of same genetic makeup ___ **identical twin.**
- ❖ tolerance to ..(**t cell**).... is longer as compared to b-cell.
- ❖ Animals with hybrid nature of Immune cells with host and donor cells are ___ **Chimeras**
- ❖ Patient suffering AIDS are susceptible to ___ **Kaposi Sarcoma**
- ❖ B-cells are activated directly without the help of T-cells for antibodies production in ___ **T- independent antigens**
- ❖ Infectious diseases that spread from person to person are said to be ___ **Contagious**
- ❖ T- Dependent antigens contain ----- type of epitopes with few copies ___ **Same**
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- ❖ Type 1 hypersensitivity is mediated by ___ **IgE**
- ❖ Pro- inflammatory cytokines are involved in ----- of inflammatory process. ___ **Up regulation**
- ❖ These all are true for immune evasion EXCEPT ----- **maximize the ability of organisms to develop infection**
- ❖ These all are true for inflammation EXCEPT ---- **Expressed in myeloid cells**
- ❖ **Adjuvant** are of ___ **2 types**
- ❖ Lymph nodes, spleen and payer's patches are. ___ **Secondary lymphoid organs**
- ❖ Second most common serum antibody... **IGA**
- ❖ Humans can make **10^7 - 10^8** different Ab molecules
- ❖ T-cells maturation in... **Thymus**
- ❖ Class switching of immunoglobulin occurs from IgM to ----- **IgG**
- ❖ Cytidine deamine is present in ----- **B cells**
- ❖ The gene for ----- chain is located on **chromosome 14** ___ **Heavy chain**
- ❖ MHC 11 molecules contain highly conserved ----- regions for binding to CD4 receptors. ___ **$\alpha 2$ & $\beta 2$**
- ❖ Antigen binding Groove of Class 11 MHC can accommodate the peptides consisting of ___ **13-25**
- ❖ Avidity in influenced by the ----- of both antigen and antibody. **Valency**
- ❖ Reaction of type 1 hypersensitivity ___ **Usually 15-30 min, sometimes 10-12 hours**
- ❖ Time required for type 111 ___ **3-10 hours**
- ❖ Th1 cells release cytokine like IFN γ upon activation which is inhibitory for ----- **Th2 cells**
- ❖ MHC stands for ___ **Major Histocompatibility Complex.**
- ❖ Symptoms of Inflammation ___ **pain, heat, redness, swelling & loss of tissue functions**
- ❖ Exudation of plasma proteins & fluids ___ **Edema**
- ❖ Which statement is true regarding t-cell and b-cell ___ **t-cell matures in thymus and b cell matures in bone marrow**
- ❖ Soluble antigens are ___ **Less immunogenic.**
- ❖ Framework Regions are Products of differentregion genes. ___ **variable**
- ❖ Least common serum antibody is ___ **IgE**
- ❖ Affinity.....**Strength of binding B/W single antigenic determinant & single antibody site.**
- ❖ Agglutination.....**Particulate nature of antigen causes the reaction of antibody to form visible clumps.**
- ❖ Unknown Ab can be determined by ___ **direct ELISA.**

- ❖ Role of IgE in allergic reactions__ **Involved in allergic reactions (Anaphylaxis)**
- ❖ Primary function of coagulation__ **Antimicrobial activity**
- ❖ How many kinds of polyadenylation__ **two**
- ❖ Example of live attenuated vaccine__ **Bacille Calmette-Guerin vaccine (BCG)**
- ❖ Time span of acute rejection__ **1-3 weeks**
- ❖ ADCC is another form of__ **Antibody dependent Cellular Cytotoxicity**
- ❖ MASP stands for ____ **Mannose-Associated Serine Proteases**